

USING PICKit™ 3 IN-CIRCUIT DEBUGGER

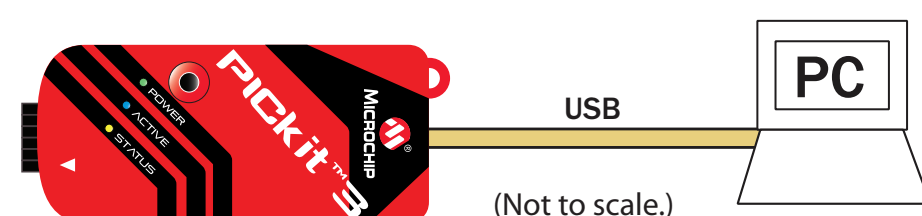
1 Install the Latest Software

Install the MPLAB® IDE software onto your PC using the MPLAB IDE CD-ROM or download the software from the MPLAB IDE page of the Microchip web site (www.microchip.com/MPLAB). Check the latest Release Notes for additional information.

2 Configure PC USB Communications

Connect the PICKit™ 3 in-circuit development debugger/programmer to a PC USB port via a USB cable. PICKit 3 uses the standard HID USB Windows® driver.

Note: If a USB hub is used, the hub must be powered with its own power supply.



3 Build Your Project

1. Launch MPLAB IDE.
2. Load your project or use the Project Wizard to create a new one.
3. Build your project based on your configurations and options.
4. Select the PICKit 3 as either a debugger (*Debugger>Select Tool>PICKit 3*) or as a programmer (*Programmer>Select Programmer>PICKit 3*).

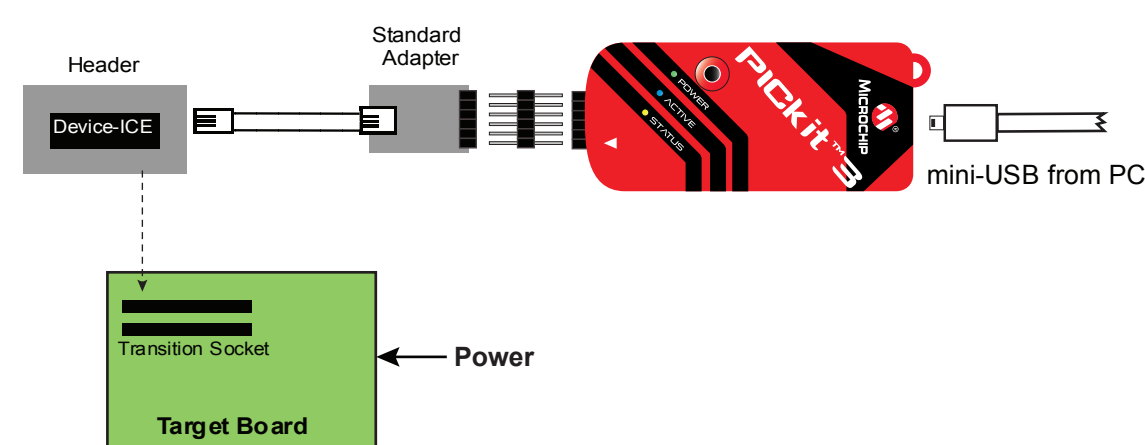
4 Connect to Target and Power

1. Attach the PICKit 3 to the PC using the USB cable, if not already.
2. Attach the communications cable between the debugger and target board.
3. Connect power to the target board.

Typical Debugger System – Device With On-Board ICE Circuitry:



Alternate Debugger System – ICE Device:



5 Program and Debug

1. Program your device.
2. As a programmer, PICKit 3 will automatically run your code. As a debugger, you can run, halt, single step and set breakpoints in your code.

Note: For information on reserved resources used by the debugger, see the PICKit 3 online help.

ADDITIONAL INFORMATION

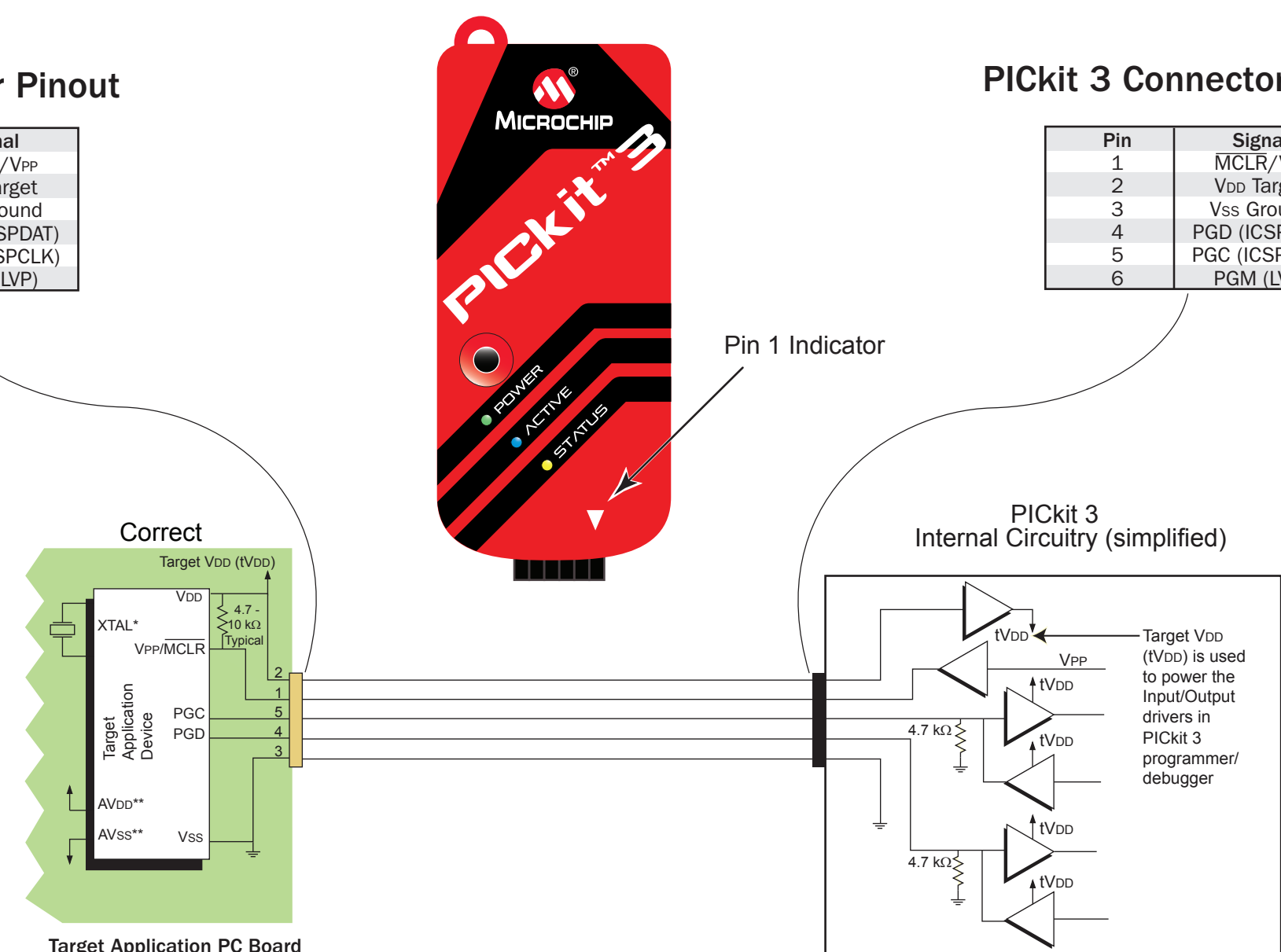
Circuitry and Connector Pinouts

Target Connector Pinout

Pin	Signal
1	MCLR/VPP
2	VDD Target
3	VSS Ground
4	PGD (ICSPDAT)
5	PGC (ICSPCLK)
6	PGM (LVP)

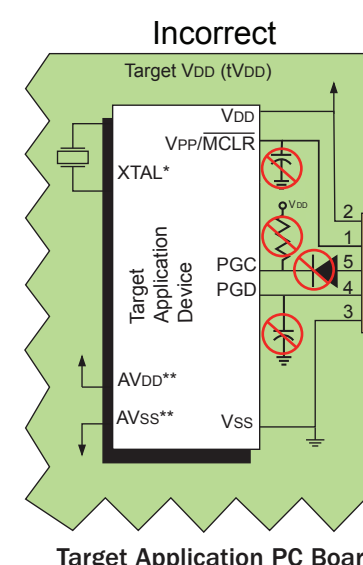
PICKit 3 Connector Pinout

Pin	Signal
1	MCLR/VPP
2	VDD Target
3	VSS Ground
4	PGD (ICSPDAT)
5	PGC (ICSPCLK)
6	PGM (LVP)



*Target device must be running with an oscillator for the debugger to function as a debugger.
**If the device has AVDD and AVSS lines, they must be connected for the debugger to operate.

Target Circuit Design Precautions



- Do not use greater than 100 µF capacitance on VDD – depending on the overall load, it will prevent the target from powering quickly when PICKit 3 is the source of power.
- Do not use capacitors on MCLR – they will prevent fast transitions of VPP.
- Do not use pull-ups on PGC/PGD – they will divide the voltage levels since these lines have 4.7 kΩ pull-down resistors in PICKit 3.
- Do not use multiplexing on PGC/PGD – they are dedicated for communications to PICKit 3.
- Do not use capacitors on PGC/PGD – they will prevent fast transitions on data and clock lines during programming and debug communications.
- Do not use diodes on PGC/PGD – they will prevent bidirectional communication between PICKit 3 and the target PIC® MCU.

Recommended Settings

COMPONENT	SETTING
Oscillator	• OSC bits set properly • Running
Power	Supplied by target
WDT	Disabled (device dependent)
Code Protect	Disabled
Table Read Protect	Disabled
LVP	Disabled
BOD	VDD > BOD VDD min
JTAG	Disabled
AVDD and AVSS	Must be connected
PGCx/PGDx	Proper channel selected, if applicable
Programming	VDD voltage levels meet programming specs

Note: See the PICKit 3 User's Guide for more component and setting information.

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